CS 4398 Software Engineering Project

Project Title: Mental Health Self-Care Tracker

Group 4
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Summary of the Project

Project Description

The Mental Health Self-Care Tracker is a desktop application that helps users build healthy routines through mood check-ins, habit tracking, journaling, and gentle reminders. The system is designed for wellness and self-care, not diagnosis or treatment. Users can record daily mood ratings, track self-care activities (sleep, hydration, exercise, meditation), set goals, and view progress trends over time.

The system supports a structured workflow: users sign in, complete daily check-ins, log habits and activities, write short journal entries, and review dashboards that summarize patterns. The application maintains an activity history and provides simple insights to encourage consistency and reflection.

Objectives / Goals

- Provide a simple, user-friendly platform for daily mood check-ins and self-care tracking.
- Help users build routines through habit goals, reminders, and streak tracking.
- Enable reflection via journaling and searchable entries.
- Display visual summaries (weekly/monthly trends) to support awareness and motivation.
- Support customization (user-defined habits, tags, and goals).
- Ensure secure access and privacy controls for sensitive personal entries.
- Deliver an MVC-based system with testable business logic and clear documentation.

Final Outcomes

By the end of the semester, the team will deliver:

- A functional JavaFX desktop application with multiple GUI screens (login, check-in, habits, journal, dashboards).
- A structured set of user workflows: check-ins, habit logging, journaling, goals, reminders, and analytics.
- Persistent storage for users, entries, habit logs, and settings.
- Unit tests for core model logic (streak calculation, goal progress, validation rules, analytics computations).

- UML artifacts (use-case diagram, class diagram, and at least one sequence diagram) aligned with the implemented system.
- Basic user documentation (how to use the app) and developer notes for maintenance.

Benefits and Limitations

Benefits

- Encourages consistent self-care routines using goals, streaks, and reminders.
- Helps users recognize patterns through dashboards and summaries.
- Provides a private space for journaling and reflection.
- Improves organization and motivation with customizable habits and tags.
- Modular design supports future extensions (mobile version, cloud sync, richer analytics).

Limitations

- The system is **not a medical tool** and does not provide diagnosis or clinical recommendations.
- Insights are informational and based on simple trend analysis (not advanced ML).
- Desktop-only (no web/mobile access in this version).
- Data sharing/integration with external health platforms is not included.

Implementation Outline

Software Development Techniques to be Used

- Agile / iterative development with 3 iterations and weekly milestones.
- MVC architecture to separate Model, View, and Controller responsibilities.
- UML-driven design (use-case, class, and sequence diagrams) to guide implementation.
- Test-driven development for core model logic and validation rules.
- Design patterns (as applicable):
    - **Observer**: UI updates when mood/habit data changes; reminder notifications.
    - **Strategy**: Different analytics calculations (weekly summary, trend computation, simple correlation).
    - **Factory**: Creation of different habit types/templates (e.g., boolean habit vs. numeric habit).

IDE, Framework, and Tools Planned for Development

- **Programming Language:** Java (JDK 17)
- **GUI Framework:** JavaFX
- **Testing:** JUnit 5

- **Build Tool:** Maven
- **Version Control:** Git + GitHub
- **UML Tool:** draw.io
- **Data Storage:** SQLite (via JDBC) *or* JSON local storage

Key Features (12 Features across 3 Iterations)

Iteration 1 — Core System & Data

1. User authentication (login/logout with secure credentials).
2. User profile management (basic preferences and personal settings).
3. Daily mood check-in (users record mood ratings and optional notes).
4. Habit management (users create and manage personal self-care habits).

Iteration 2 — Tracking & Reflection

5. Habit logging (record daily completion of self-care activities).
6. Journal entries (users write and store personal reflections).
7. Search and filter journal entries (by date or keyword).
8. Goal setting (users define weekly targets for habits).

Iteration 3 — Insights & Visualization

9. Streak tracking (track consecutive days of habit completion).
10. Dashboard overview (display mood trends and habit summaries).
11. Weekly summary report (summaries of mood averages and habit completion).
12. Reminder notifications (in-app reminders for check-ins and habits).

Note: MH-SCT is a wellness/self-care tracking tool and is not intended for diagnosis or treatment.